

Module - IV - Transient Analysis

①

Transient State :-

If a network containing energy storing elements with change in excitation, the currents and voltages change from one state to other state. The behaviour of the voltage (or) current, when it is changed from one state to another is called the transient state.

Natural Response:-

A circuit containing storage elements which are independent of the sources, the response depends upon the nature of the circuit and is called the natural response.

Transient Response:-

Energy storing elements like inductor and capacitor deliver their energy to the resistors. Hence the response changes with time, gets saturated after some time and is referred to as the transient response.

Steady state:-

A circuit consisting of constant sources is said to be in steady state if voltages and currents do not change with time.

Initial condition:-

Initial condition of a network are the values of currents and voltage before switching. The instant just before switching is denoted by $t = 0$.

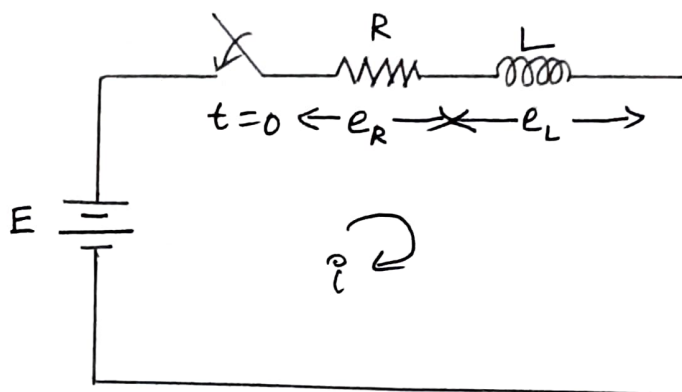
Transient time:-

The time taken for the circuit to change from one steady state to another steady state is called the transient time.

RL Transients:-

An RL circuit is connected to a battery through a switch S . Assume that the switch is closed at time $t = 0$ and also assume that the time of switching the current is zero.

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Applying KVL for the loop $E = e_R + e_L$

Voltage across the resistor, $e_R = iR$

Voltage across the inductor, $e_L = L \frac{di}{dt}$

$$E = iR + L \frac{di}{dt}$$

Laplace,

$$R I(s) + L [s I(s) - i(0)] = \frac{E}{s}$$

Initially $i(0) = 0$

$$R I(s) + L s I(s) = \frac{E}{s}$$

$$I(s) [R + sL] = E/s$$

$$I(s) = \frac{E}{s(R + sL)}$$

$$\frac{E}{s(R + sL)} = \frac{A}{s} + \frac{B}{R + sL}$$

$$\frac{E}{s(R + sL)} = \frac{A(R + sL) + Bs}{s(R + sL)}$$

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$$E = A(R + sL) + Bs$$

$$s = 0,$$

$$E = AR \Rightarrow \boxed{A = E/R}$$

$$s = -R/L,$$

$$E = \frac{-BR}{L}, \quad \boxed{B = \frac{-EL}{R}}$$

$$I(s) = \frac{E/R}{s} - \frac{EL/R}{R + sL}$$

$$I(s) = \frac{E}{Rs} - \frac{EL}{R(R + sL)}$$

$$I(s) = \frac{E}{Rs} - \frac{EL}{RL[s + R/L]} \Rightarrow \frac{E}{Rs} - \frac{E}{R(s + R/L)}$$

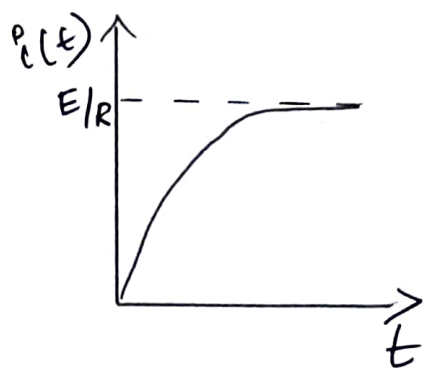
Taking Inverse Laplace transform,

$$L^{-1}[I(s)] = \frac{E}{R} L^{-1}\left(\frac{1}{s}\right) - \frac{E}{R} L^{-1}\left(\frac{1}{s + R/L}\right)$$

$$i(t) = \frac{E}{R} - \frac{E}{R} e^{-\frac{R}{L}t}$$

$$\boxed{i(t) = \frac{E}{R} \left(1 - e^{-\frac{R}{L}t}\right)}$$

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The final value of current,

$$i(t) \Big|_{t=\infty} = \frac{E}{R}$$

To determine the current at any time t , after the switch has been closed, $t = t$,

$$t = \tau = L/R$$

$$i(t) \Big|_{t=L/R} = \frac{E}{R} \left(1 - e^{-\frac{R}{L} \times \frac{L}{R}} \right)$$

$$i(t) = \frac{E}{R} (1 - e^{-1}) \quad (e^{-1} = 0.3678)$$

$$i(t) = 0.632 \frac{E}{R}$$

$i(t) = 63.2\%$ of the steady state
(or)
final value

The time constant of an RL series circuit is defined as the time during which the current increases to 63.2% of its steady state value.

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The transient voltage across the resistance

$$e_R = iR$$

$$e_R = \frac{E}{R} \left(1 - e^{-R/Lt}\right) R$$

$$e_R = E(1 - e^{-R/Lt})$$

The transient voltage across the inductor

$$e_L = L \frac{di}{dt}$$

$$e_L = L \frac{d}{dt} \left[\frac{E}{R} (1 - e^{-(R/L)t}) \right]$$

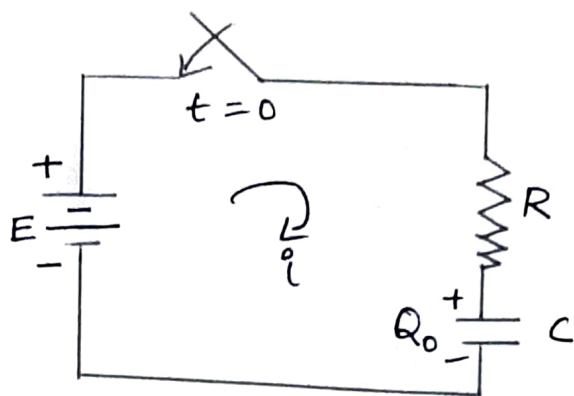
$$e_L = \frac{E}{R} L \frac{d}{dt} [1 - e^{-(R/L)t}]$$

$$e_L = \frac{EL}{R} \left[-e^{-(R/L)t} \left[\frac{-R}{L} \right] \right]$$

$$e_L = E e^{-(R/L)t}$$

RC Transient:-

consider the RC series circuit with a DC voltage applied through a switch. Let the capacitance have an initial charge of Q_0 coulombs,



The initial voltage on the capacitor, $V_0 = \frac{Q_0}{C}$

Applying KVL, we get

$$iR + \frac{1}{C} \int i dt + V_0 = E$$

If there is no initial charge $Q_0 = 0$ (i.e.) $V_0 = 0$

$$iR + \frac{1}{C} \int i dt = E$$

Using Laplace Transformation,

$$R I(s) + \frac{1}{C} \frac{I(s)}{s} = \frac{E}{s}$$

$$\left(R + \frac{1}{sC}\right) I(s) = \frac{E}{s}$$

$$I(s) = \frac{E}{s\left(R + \frac{1}{sC}\right)}$$

$$I(s) = \frac{E}{sR + \frac{1}{C}}$$

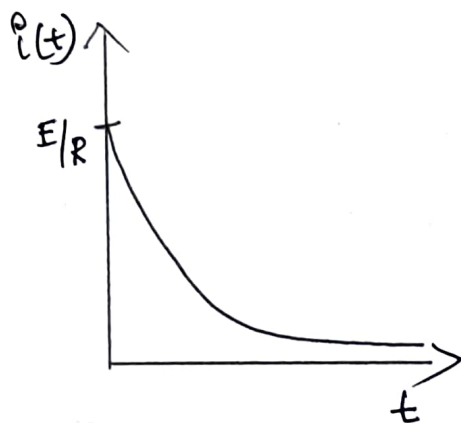
$$I(s) = \frac{E}{R\left(s + \frac{1}{RC}\right)}$$

Taking inverse Laplace transform on both sides,

$$i(t) = \frac{E}{R} e^{-t/RC}$$

$$i(t) \Big|_{t=0} = \frac{E}{R}$$

$$i(t) \Big|_{t=\infty} = 0$$



The initial value of the current is $\frac{E}{R}$ amperes and the final steady state value is zero ampere.

Voltage across the resistor, $e_R = iR = E e^{-t/RC}$

Voltage across the capacitor, $e_C = \frac{1}{C} \int_0^t i(t) dt$

$$e_C = \frac{E}{CR} \int_0^t e^{-t/RC} dt$$

$$e_C = \frac{E}{CR} \left[\frac{-e^{-t/RC}}{1/RC} \right]_0^t$$

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$$e_c = -E \left(e^{-t/RC} \right)_0^t$$

$$e_c = -E \left(e^{-t/RC} - 1 \right)$$

$$e_c = E \left(1 - e^{-t/RC} \right)$$

e_c increases with time when $t = \infty$ (steady state)
 e_c is equal to the applied voltage and opposing it. The current i then becomes zero.

Time constant $\tau = RC$ seconds

$$P_R = e_R i = \frac{E^2}{R} e^{-2t/RC}$$

$$P_C = e_c i = E \left(1 - e^{-t/RC} \right) \frac{E}{R} \left(e^{-t/RC} \right)$$

$$P_C = \frac{E^2}{R} \left(e^{-t/RC} - e^{-2t/RC} \right)$$

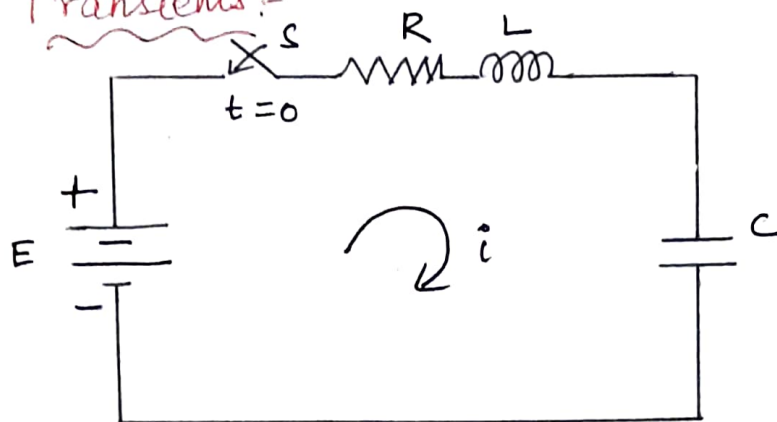
The energy stored in the capacitor is obtained by integrating P_C from 0 to ∞ .

$$E_C = \frac{E^2}{R} \int_0^{\infty} \left(e^{-t/RC} - e^{-2t/RC} \right) dt$$

$$E_C = \frac{E^2}{R} \left[\frac{e^{-t/RC}}{-1/RC} + \frac{e^{-2t/RC}}{2/RC} \right]_0^{\infty}$$

$$E_C = \frac{E^2}{R} \times \frac{RC}{2} \Rightarrow \underline{\underline{E_C = \frac{1}{2} C E^2}}$$

At $t = RC$, $i = E/R e^{-1} \Rightarrow \underline{\underline{E_C = 36.8\% \text{ of the initial current}}}$

RLC Transients:-

Applying KVL to the series RLC circuit

$$iR + L \frac{di}{dt} + \frac{1}{C} \int i dt = E$$

Assume no initial charge on the capacitor

Taking Laplace transform on both sides,

$$RI(s) + L[sI(s) - i(0)] + \frac{1}{C} \frac{I(s)}{s} = \frac{E}{s}$$

Assuming $i(0) = 0$ we get,

$$RI(s) + LsI(s) + \frac{I(s)}{Cs} = \frac{E}{s}$$

$$I(s) \left[R + sL + \frac{1}{Cs} \right] = \frac{E}{s}$$

$$I(s) = \frac{E}{s(R + sL + 1/Cs)}$$

$$I(s) = \frac{E}{(Rs + s^2L + 1/C)}$$

$$I(s) = \frac{E/L}{s^2 + R/Ls + 1/LC}$$

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The roots of the denominator are

$$s = \frac{-R/L \pm \sqrt{(R/L)^2 - 4/LC}}{2}$$

$$s = \frac{-R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - 1/LC}$$

$$s = \alpha \pm \beta$$

$$\alpha = \frac{-R}{2L}, \beta = \sqrt{\left(\frac{R}{2L}\right)^2 - 1/LC}$$

case (i) Discriminant positive

$$\left(\frac{R}{2L}\right)^2 > 1/LC$$

The two roots are real and distinct

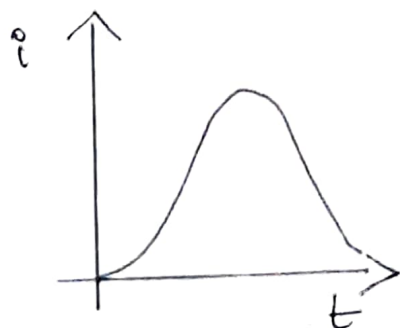
$$s_1 = \alpha + \beta, s_2 = \alpha - \beta$$

$$I(s) = \frac{K_1}{s - (\alpha + \beta)} + \frac{K_2}{s - (\alpha - \beta)}$$

$$i(t) = K_1 e^{(\alpha + \beta)t} + K_2 e^{(\alpha - \beta)t}$$

$$i(t) = e^{\alpha t} (K_1 e^{\beta t} + K_2 e^{-\beta t})$$

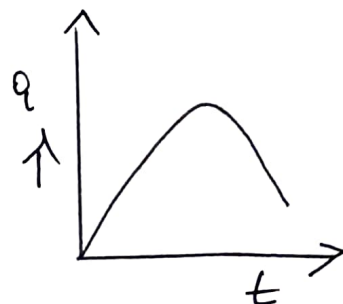
The current is over damped.



case (ii) Discriminant zero

$$\left(\frac{R}{2L}\right)^2 - \frac{1}{LC} = 0$$

$$\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$$



The two roots are equal

$$I(s) = \frac{K_1}{(s-\alpha)^2} + \frac{K_2}{(s-\alpha)}$$

$$i(t) = K_1 t e^{\alpha t} + K_2 e^{\alpha t}$$

$$i(t) = e^{\alpha t} (K_1 t + K_2)$$

The current is critically damped.

case (iii) Discriminant negative

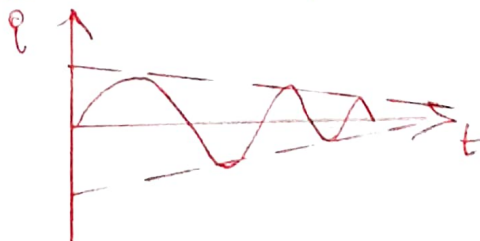
$$\left(\frac{R}{2L}\right)^2 < \frac{1}{LC}$$

The roots are complex conjugate.

$$I(s) = \frac{K_1}{s - (\alpha + j\beta)} + \frac{K_2}{s - (\alpha - j\beta)}$$

$$i(t) = e^{\alpha t} [K_1 e^{j\beta t} + K_2 e^{-j\beta t}]$$

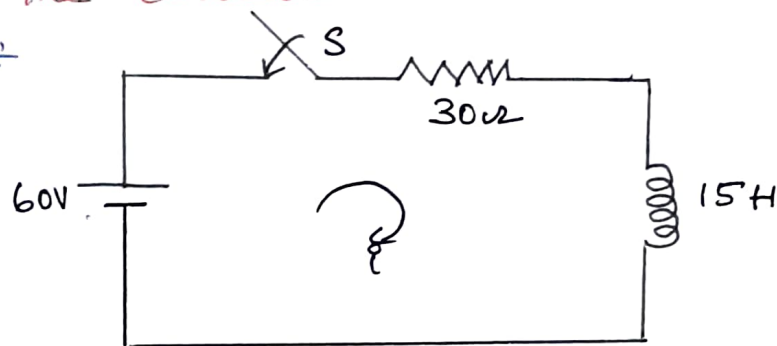
The current is oscillatory in nature and the $\alpha = -R/2L$ is always negative.



Problem : 1

A series RL circuit with $R = 30\Omega$ and $L = 15H$ has a constant voltage $E = 60V$ applied at $t = 0$ as shown in figure. Determine the current i , the voltage across resistor and the voltage across the inductor.

Solution:-



Applying KVL, we get

$$30i + 15 \frac{di}{dt} = 60$$

$$2i + \frac{di}{dt} = 4$$

Taking Laplace transform,

$$2I(s) + [sI(s) - i(0)] = \frac{4}{s}$$

Assuming zero initial conditions

$$2I(s) + sI(s) = \frac{4}{s}$$

$$I(s) [s + 2] = \frac{4}{s}$$

$$I(s) = \frac{4}{s(s+2)}$$

Taking partial fractions,

$$\frac{4}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2}$$

Put $s = -2$

$$\frac{4}{s(s+2)} = \frac{A(s+2) + Bs}{s(s+2)}$$

$$4 = A(-2+2) + B(-2)$$

$$-2B = 4$$

$$B = 4/-2 = -2$$

Put $s = 0$

$$4 = A(0+2) + B(0)$$

$$2A = 4$$

$$A = 4/2 = 2$$

$$I(s) = \frac{2}{s} - \frac{2}{s+2}$$

Taking inverse laplace transform,

$$L^{-1}[I(s)] = 2L^{-1}(1/s) - 2L^{-1}\left(\frac{1}{s+2}\right)$$

$$\underline{i(t) = 2 - e^{-2t} = 2(1 - e^{-2t}) \text{ Amps}}$$

Voltage across resistor = iR

$$e_R = 2(1 - e^{-2t}) \times 30$$

$$\underline{e_R = 60(1 - e^{-2t}) \text{ volts}}$$

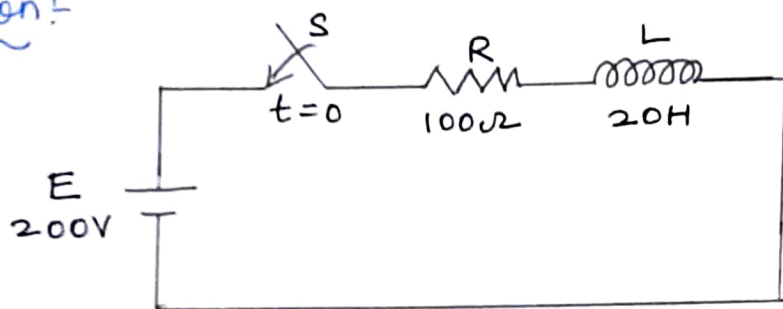
voltage across inductor, $e_L = L \frac{di}{dt}$

$$e_L = 15 \frac{d}{dt} 2(1 - e^{-2t}) = 30 \times 2e^{-2t}$$

$$\underline{e_L = 60e^{-2t} \text{ volts}}$$

Problem: 2

A series RL circuit with $R = 100\Omega$ and $L = 20H$ has a DC voltage of $200V$ applied through a switch at $t = 0$. Find (1) the equation for the current and voltage across the different elements (2) the current at $t = 0.5$ seconds (3) the current at 1 sec and (4) the time at which $e_R = e_L$.

Solution:-

$$\text{Current, } i(t) = \frac{E}{R} (1 - e^{-(R/L)t})$$

$$i(t) = \frac{200}{100} (1 - e^{-(\frac{100}{20})t})$$

$$\underline{i(t) = 2(1 - e^{-5t}) \text{ A}}$$

$$\text{voltage across the resistance, } e_R = E (1 - e^{-(R/L)t})$$

$$e_R = 200 (1 - e^{-(\frac{100}{20})t})$$

$$\underline{e_R = 200(1 - e^{-5t}) \text{ volts}}$$

$$\text{voltage across the inductance, } e_L = E e^{-(R/L)t}$$

$$e_L = 200 e^{-(\frac{100}{20})t} \Rightarrow \underline{e_L = 200 e^{-5t} \text{ volts}}$$

$$\text{current at } t = 0.5 \text{ seconds, } i(0.5) = 2(1 - e^{(-5 \times 0.5)}) \text{ A}$$

$$\underline{i(0.5) = 1.84 \text{ A}}$$

current at $t=1$ seconds,

$$i(1) = 2(1 - e^{(-1/0.2)})$$

$$\underline{i(1) = 1.99 \text{ A}}$$

Time when $e_R = e_L$

$$200(1 - e^{-5t}) = 200e^{-5t}$$

$$1 = e^{-5t} + e^{-5t}$$

$$2e^{-5t} = 1$$

$$e^{-5t} = 1/2$$

Taking natural logarithm on both sides

$$\ln(e^{-5t}) = \ln(1/2)$$

$$-5t = -\ln(2)$$

$$t = \frac{\ln(2)}{5}$$

$$\underline{t = 0.1386 \text{ sec}}$$

Problem:-3

An RLC series circuit has $R=10\Omega$, $L=2\text{ H}$. what value of capacitance will make the circuit critically damped.

Solution:-

For critical damping, $\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$

$$C = \frac{4L}{R^2} = \frac{4 \times 2}{10^2} \Rightarrow \underline{C = 0.08 \text{ F}}$$

Problem: 4

A series RLC circuit has $R=20\Omega$, $L=0.1H$ and $C=100\mu F$. Determine whether the circuit is overdamped, critically damped (or) under damped.

Solution:-

$$\text{Damping factor, } \alpha = \frac{R}{2L}$$

$$\alpha = \frac{20}{2 \times 0.1} = 100$$

$$\text{Undamped natural frequency, } \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{0.1 \times 100 \times 10^{-6}}}$$

$$\omega_0 = 316.228 \text{ rad/sec}$$

Comparing α & $\omega_0 \Rightarrow 100 < 316.228$, $\alpha < \omega_0$ so the circuit is underdamped system.

Problem: 5

Find the time constant of RL circuit having $R=10\Omega$ and $L=0.1mH$.

Solution:-

$$\text{Time constant, } \tau = L/R$$

$$\tau = \frac{0.1 \times 10^{-3}}{10}$$

$$\underline{\underline{\tau = 10\mu s}}$$

Problem: 6

A DC voltage of 100 volts is applied to a series RL circuit with $R = 25\Omega$. What will be the current in the circuit at twice the time constant.

Solution:-

$$E = 100V, R = 25\Omega$$

$$i(t) = \frac{E}{R} (1 - e^{-R/L t})$$

$$\text{where, } \tau = L/R$$

$$i(t) = \frac{E}{R} (1 - e^{-t/\tau})$$

$$\text{given } t = 2\tau$$

$$i(t) = \frac{100}{25} (1 - e^{-2\tau/\tau})$$

$$i(t) = 4(1 - e^{-2})$$

$$\underline{\underline{i(t) = 3.45A}}$$

Problem: 7

Find the time constant of RC circuit having $R = 26\Omega$ and $C = 8\mu F$.

Solution:-

$$\text{Time constant, } \tau = RC$$

$$\tau = 26 \times 8 \times 10^{-6}$$

$$\underline{\underline{\tau = 208\mu s}}$$

Problem :- 8

A series RC with $R = 5 \text{ k}\Omega$ and $C = 20 \mu\text{F}$ has a constant voltage source of 100 V applied at $t = 0$, the capacitor is initially uncharged. Derive the expression for current $i(t)$ and the charge $q(t)$ for $t > 0$ and analyze how the current and voltage across the capacitor vary with time.

Solution :-

capacitor initially uncharged, so $q(0) = 0$

$$\text{and } i(0) = E/R$$

$$\text{where, } q(t) = q_f (1 - e^{-t/RC})$$

$$q_f = CV \text{ (Final steady state charge)}$$

$$q(t) = 20 \times 10^{-6} \times 100$$

$$\underline{q(t) = 2 \times 10^{-3} (1 - e^{-t/0.1})}$$

$$i(t) = \frac{E}{R} e^{-t/RC}$$

$$= \frac{100}{5000} e^{-t/0.1} \text{ A}$$

$$\underline{i(t) = 0.02 e^{-t/0.1} \text{ A}}$$

$$e_c = E(1 - e^{-t/RC}) \Rightarrow 100(1 - e^{-t/0.1}) \text{ V}$$

$$\underline{e_c = 100(1 - e^{-t/0.1}) \text{ V}}$$

Problem:- 9

A street-lighting system uses a small RL circuit to control how quickly the lamps reach full brightness when switched on. The circuit consists of a coil with resistance $R=20\Omega$ and inductance $L=4H$ connected to a $24V$ DC supply through a switch. Assume the current through the coil is zero before switching. (i) Analyze the circuit and derive $i(t)$ when the switch is closed at $t=0$ (ii) calculate the time constant at $t=0.2s$ and steady state current (iii) Explain in practical terms, what the transient behaviour means for the lamp brightness (iv) sketch the waveform of current versus time and mark $t=0, \tau, 3\tau, 5\tau$.

Solution:-

$$i(t) = \frac{E}{R} (1 - e^{-(R/L)t})$$

$$E=24V, R=20\Omega, L=4H \quad - \left(\frac{20}{4}\right)t$$

$$i(t) = \frac{24}{20} (1 - e^{-(\frac{20}{4})t})$$

$$i(t) = 1.2 (1 - e^{-5t})$$

where, $\tau = L/R$

$$\tau = 4/20 = 0.2$$

$$i(t) = 1.2 (1 - e^{-t/\tau})$$

When $t=0.2s$ & $\tau=0.2 \Rightarrow t=\tau$

$$i(0.2) = 1.2 (1 - e^{(-0.2/0.2)})$$

$$i(0.2) = 0.758 A$$

steady state current, $i(t) = E/R$

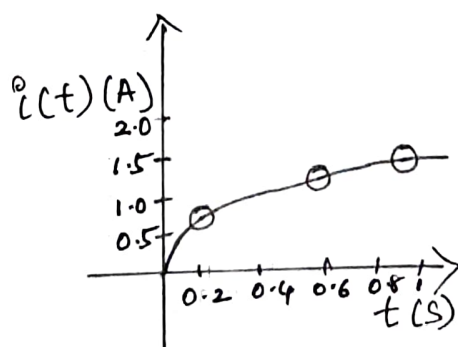
$$i(t) = \frac{24}{20}$$

$$i(t) = 1.2 \text{ A}$$

Transient Analysis and lamp brightness depends upon the switch position and the flow of current in the circuit. At $t=0$, current starts from zero and increases exponentially towards its final value, $i(t) = 1.2 \text{ A}$. The lamp brightness gradually increases and when $t = 5\tau = 1 \text{ second}$ the circuit reaches steady state and the lamp brightness is 99%.

Output response curve:-

Time (Seconds)	$i(t) \text{ (A)}$
$t = 0$	0
$t = \tau = 0.2 \text{ s}$	0.76
$t = 3\tau = 0.6 \text{ s}$	1.14
$t = 5\tau = 1.0 \text{ s}$	1.19



Problem:-10

A $50\mu\text{F}$ capacitor is discharged through a $100\text{k}\Omega$ resistor. If the capacitor is initially charged to 400V , determine the initial energy stored in the capacitor and the current at time $t=5\text{s}$.

Solution:-

RC series circuit, $R=100\text{k}\Omega$, $C=50\mu\text{F}$
and $E=400\text{V}$.

$$i(t) = \frac{E}{R} e^{-t/RC}$$

$$i(t) = \frac{400}{100 \times 10^3} e^{-t/(100 \times 10^3 \times 50 \times 10^{-6})}$$

$$\underline{i(t) = 0.004 e^{-t/5}}$$

When, $t=5\text{s}$

$$i(5) = 0.004 e^{-5/5}$$

$$\underline{i(5) = 1.47\text{mA}}$$

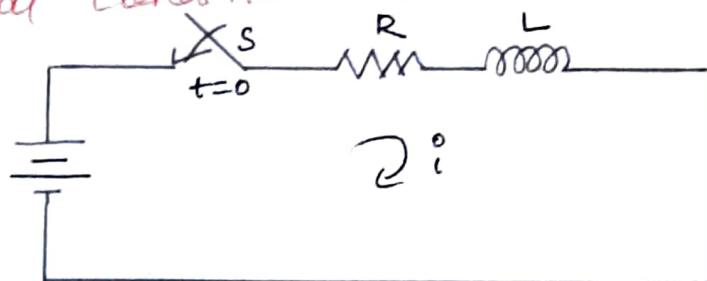
Energy stored in capacitor, $E_c = \frac{1}{2} C E^2$

$$E_c = \frac{1}{2} \times (50 \times 10^{-6}) \times (400)^2$$

$$E_c = 4\text{J}$$

Initial energy stored
in capacitor } $E_c = 4\text{J}$

Derive the expression for current i in RL transient response with impulse input at initial condition $i(0) = 0$.



The circuit consist of a resistor (R), an Inductor (L) and an impulse voltage source $\delta(t)$ connected in series.

Differential equation,

$$L \frac{di(t)}{dt} + Ri(t) = \delta(t)$$

where, $\delta(t) \rightarrow$ Delta function representing the impulse input

Laplace transform,

$$L[sI(s) - i(0)] + RI(s) = 1$$

$$i(0) = 0, \quad L[sI(s) - 0] + RI(s) = 1$$

$$I(s)[sL + R] = 1$$

$$I(s) = \frac{1}{sL + R}$$

$$I(s) = \frac{1}{L(s + R/L)}$$

$$I(s) = \frac{1/L}{s + R/L}$$

Applying inverse Laplace transform on both side,

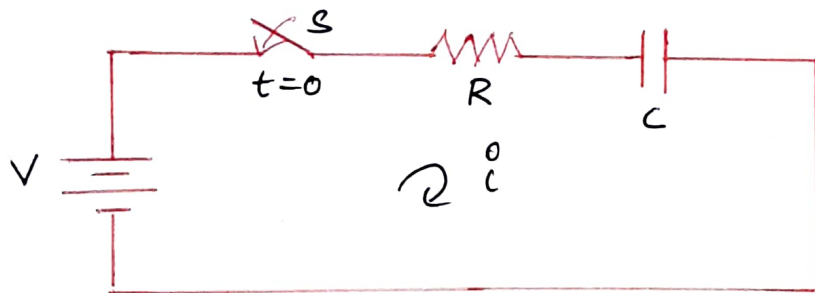
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$$L^{-1}[I(s)] = L^{-1}\left[\frac{1/L}{s + R/L}\right]$$

$$\underline{\underline{i(t) = \frac{1}{L} e^{-(R/L)t}}}$$

This expression is valid for $t > 0$. The constant $\tau = L/R$ is the time constant of the RL circuit.

Derive the expression for current in RC transient response with impulse input at initial condition $i(0) = 0$.



The circuit consists of a resistor (R) and a capacitor (C) connected in series with an impulse voltage source $\delta(t)$. The initial condition is that the capacitor is initially uncharged, so the voltage across the capacitor $V_C(0)$ is 0V, this implies that $i(0) = 0$ A.

Applying KVL,

$$V_R(t) + V_C(t) = V_S(t)$$

$$R i(t) + V_C(t) = \delta(t)$$

$$R i(t) + \frac{1}{C} \int i(t) dt = \delta(t)$$

Applying Laplace Transform,

$$L[R i(t) + \frac{1}{C} \int i(t) dt] = L[\delta(t)]$$

$$R I(s) + \frac{1}{C} \left(\frac{I(s)}{s} \right) = 1$$

$$I(s) \left[R + \frac{1}{sC} \right] = 1$$

$$I(s) \left[\frac{RCs + 1}{sC} \right] = 1 \Rightarrow I(s) = \frac{Cs}{RCs + 1}$$

÷ by RC

$$I(s) = \frac{sC/RC}{\frac{RCs + 1}{RC}} \Rightarrow \frac{1}{R} - \frac{1}{R^2C} \frac{1}{s + 1/RC}$$

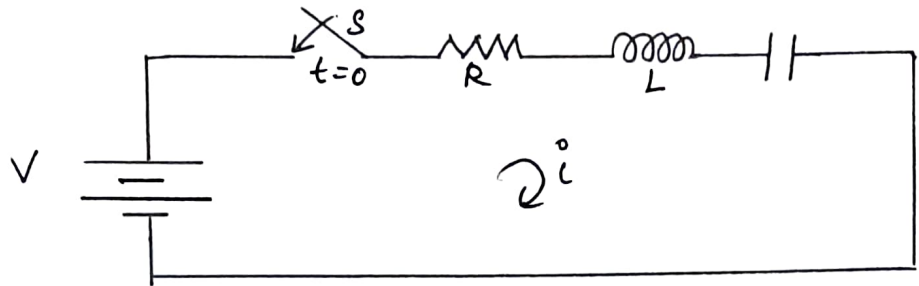
Taking Inverse Laplace Transform,

$$i(t) = L^{-1} \left[\frac{1}{R} - \frac{1}{R^2C} \frac{1}{s + 1/RC} \right]$$

$$i(t) = \frac{1}{R} \delta(t) - \frac{1}{R^2C} e^{-(t/RC)} u(t)$$

The current has an impulse component at $t=0$ and an exponential decay for $t > 0$.

Derive the expression for current in RLC transient response with impulse input at initial condition $i(0)=0$.



The circuit consists of a resistor (R), an inductor (L) and a capacitor (C) connected in series with an impulse voltage source $\delta(t)$. The initial conditions are $i(0)=0$ and the capacitor is uncharged.

Applying KVL,

$$V_R(t) + V_L(t) + V_C(t) = V_S(t)$$

$$Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt = \delta(t)$$

Applying Laplace Transform,

$$RI(s) + LsI(s) + \frac{1}{sC} I(s) = 1$$

$$I(s) \left[R + Ls + \frac{1}{sC} \right] = 1$$

$$I(s) \left[\frac{sRC + s^2LC + 1}{sC} \right] = 1$$

$$I(s) = \frac{sC}{s^2LC + sRC + 1}$$

$$I(s) = \frac{s/L}{s^2 + s(R/L) + 1/(LC)}$$

The Inverse Laplace transform of $I(s)$ depends on the damping of the circuit, which is determined by the roots of the denominator polynomial, $s^2 + s(R/L) + 1/LC = 0$. The roots are given by the quadratic formula.

$$\text{Underdamped} \Rightarrow \left(\frac{R}{2L}\right)^2 < 1/LC$$

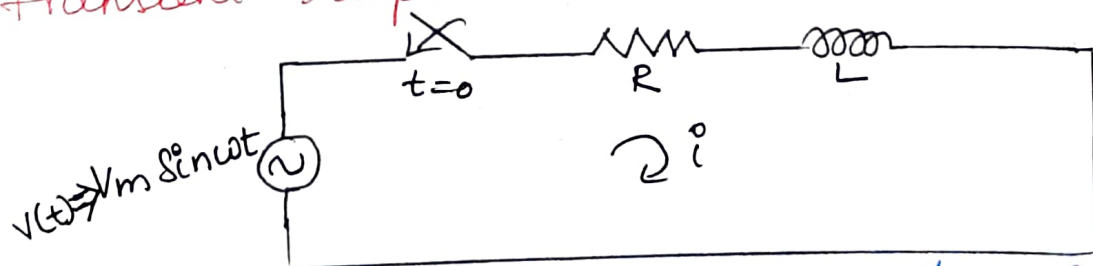
$$\text{Critically damped} \Rightarrow \left(\frac{R}{2L}\right)^2 = 1/LC$$

$$\text{Overdamped} \Rightarrow \left(\frac{R}{2L}\right)^2 > 1/LC$$

The final expression for $i(t)$ vary depending on the above three cases. For a general solution, the Inverse Laplace transform gives,

$$\underline{i(t) = \mathcal{L}^{-1} \left[\frac{s/L}{s^2 + s(R/L) + 1/LC} \right]}$$

Derive the expression for current in RL transient response with sinusoidal input.



The circuit consist of $R \times L$ connected in series and a sinusoidal voltage source $V(t) = V_m \sin(\omega t)$. Initial condition, $t=0 \Rightarrow i(0) = 0$.

Applying KVL,

$$V_R(t) + V_L(t) = V(t)$$

$$R i(t) + L \frac{di(t)}{dt} = V_m \sin(\omega t)$$

General solution for first order, non-homogeneous linear equation is given by,

$$i(t) = i_h(t) + i_p(t)$$

$$i_h(t) = A e^{-(R/L)t}$$

$$i_p(t) = I_m \sin(\omega t + \phi)$$

$$\text{Where, } I_m = \frac{V_m}{|Z|} = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}}$$

current lags the voltage by the phase angle ϕ , $\phi = -\theta = -\tan^{-1}(\omega L/R)$

$$i(t) = A e^{-(R/L)t} + \frac{V_m}{Z} \sin(\omega t + \phi)$$

$$i(0) = 0, \quad i(0) = A e^0 + \frac{V_m}{Z} \sin(0 + \phi) = 0$$

$$\begin{aligned} A + \frac{V_m}{Z} \sin(\phi) &= 0 \\ A &= -\frac{V_m}{Z} \sin(\phi) \end{aligned} \quad \left| \begin{array}{l} \text{where,} \\ Z = \sqrt{R^2 + (\omega L)^2} \\ \phi = \tan^{-1}(\omega L/R) \\ \tau = L/R \end{array} \right.$$

$$i(t) = -\frac{V_m}{Z} \sin(\phi) e^{-(R/L)t} + \frac{V_m}{Z} \sin(\omega t + \phi)$$

$$i(t) = \frac{V_m}{Z} [\sin(\omega t + \phi) - \sin(\phi) e^{-(R/L)t}] \quad \text{for } t \geq 0$$